

Lab – 2 Bootstrap

Step 1: Bootstrap CDN (Connecting Bootstrap)

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
```

What is this?

This connects your page to **Bootstrap 5 CSS library**.

Why needed?

Without this:

- .container won't work
- .d-flex won't work
- .text-center won't work
- All Bootstrap styling will fail

Think of it as:

“Importing Bootstrap brain into your HTML”

Step 2: .container (Layout Wrapper)

```
<div class="container">
```

What it does:

- Creates a responsive layout area
- Adds left & right padding automatically
- Keeps content centered

Teaching Tip:

There are 3 main containers:

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Class	Behavior
container	Fixed width responsive
container-fluid	Full width (100%)
container-sm/md/lg	Responsive by breakpoint

In your code:

```
container d-flex justify-content-center align-items-center vh-100
```

This container also handles centering.

Step 3: Flexbox Utilities

d-flex

justify-content-center

align-items-center

vh-100

These are **Bootstrap Utility Classes**.

- **d-flex**

Means:

display: flex;

Turns the container into a **flexbox layout**.

Without this → alignment utilities won't work.

- **justify-content-center**

Centers horizontally.

Works on the X-axis.

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- **align-items-center**

Centers vertically.

Works on the Y-axis.

- **vh-100**

Means:

height: 100vh;

Makes container full screen height.

So together:

.container d-flex justify-content-center align-items-center vh-100

= Perfect center of screen (horizontal + vertical)

Step 4: Typography Utilities

Used inside body:

fw-bold

text-muted

text-center

text-start

- **fw-bold**

Makes text bold.

Equivalent to:

font-weight: bold;

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- **text-muted**

Makes text gray/light color.

Used for subtitles or less important text.

- **text-center**

Aligns text center.

- **text-start**

Aligns text left (start of line).

- **Step 5: Spacing Utilities**

Bootstrap uses a spacing scale:

0 → no space

1 → small

2 → medium small

3 → medium

4 → large

5 → extra large

In your code:

mb-0

mb-2

mb-3

- **mb-3**

Margin Bottom = medium space

- **mb-2**

Margin Bottom = smaller space

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Step 6: Image Styling with Bootstrap

`class="student-img mb-3"`

Only `mb-3` is Bootstrap.

Adds spacing below image.

Step 7: Viewport Meta (Responsive Important)

`<meta name="viewport" content="width=device-width, initial-scale=1">`

Why important?

Without this:

- Mobile view will break
- Bootstrap responsiveness won't work properly

This is mandatory for responsive design.

Step 8: Horizontal Rule `<hr>`

`<hr>`

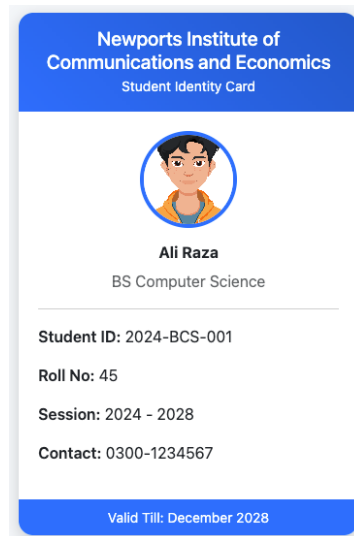
Bootstrap automatically styles `<hr>`.

It gives a clean separator line.

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Class – Task:

Create the Student ID card using HTML + Bootstrap



Hint: follow the anatomy of Elements

